

Aegis

Privacy-Preserving Clinical AI with Cryptographic Consent
and Verifiable Proof-of-Inference

Frontier T

The Problem

Healthcare AI is extractive by default: patients cannot see who accessed their data, under what authorization, or with what model. Outputs may hallucinate with no way to verify fidelity. There is no tamper-proof audit trail or per-use accountability.

Our Solution

Aegis combines local AI inference with on-chain consent, provenance, and settlement. AI runs on the document locally; only hashes and metadata go on-chain (never raw PHI). Each admissible extraction mints a Proof-of-Inference receipt, settled per use in stablecoin.

How It Works

- Patient grants scoped consent on-chain

- Provider

End Product

- Web App (Next.js): Patient, Provider, Verify, Reconciliation, Showcase

- AI Servi

Why AI + Blockchain Together

Removing AI leaves a dead consent registry with no extraction utility. Removing blockchain makes outputs unprovable with no audit trail or settlement. Both halves are essential.

Privacy and Hackathon Fit

- No PHI on-chain - hashes, Merkle roots, and metadata only

- Track: H

One-Sentence Summary

Aegis gives patients cryptographic control over clinical AI access, runs extraction locally with measurable anti-hallucination grounding, and leaves a publicly verifiable on-chain receipt and payment record for every authorized inference - without putting patient data on the blockchain.

Tech stack: Next.js, FastAPI, Ollama, Solidity, Base Sepolia. See SPEC.md and README.md in the project repository for full technical detail.